

MINCHUMINA, AK, USA

WMO: 702460

Lat: 63.886N

Lon: 152.302W

Elev: 678

StdP: 14.34

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26512

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.5	-32.6	-29.3	1.1	-17.4	-28.2	1.2	-16.0	15.4	5.7	12.7	2.5	0.1	30	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	77.5	59.2	73.4	57.6	71.6	56.8	61.4	73.1	59.7	70.4	58.2	67.8	3.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.5	69.8	64.4	54.8	65.6	62.8	53.8	63.2	62.3	27.7	73.4	26.5	70.1	25.3	67.8	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	10.7	8.7	-38.6	81.7	10.9	4.1	-46.4	84.6	-52.8	87.0	-59.0	89.3	-66.9	92.2		
(5)			-30.7	63.8	3.1	2.5	-32.9	65.6	-34.7	67.1	-36.4	68.5	-38.7	70.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	-3.7	3.5	11.4	30.6	47.5	57.9	60.0	54.8	45.1	26.2	6.8	-1.0
	DBStd	25.41	15.75	13.10	11.59	10.08	7.81	5.93	5.44	5.81	6.83	10.04	11.46	12.78
	HDD50	8695	1665	1301	1198	583	138	6	3	17	170	738	1296	1580
	HDD65	13398	2130	1721	1663	1032	544	226	170	320	598	1203	1746	2045
	CDD50	803	0	0	0	1	59	242	312	167	22	0	0	0
	CDD65	32	0	0	0	0	0	12	15	5	0	0	0	0
	CDH74	296	0	0	0	0	13	116	130	37	0	0	0	0
	CDH80	32	0	0	0	0	0	16	13	3	0	0	0	0
	WSAvg	2.1	1.6	1.5	2.2	2.5	2.9	2.6	2.8	2.2	1.9	1.7	1.4	1.3
	PrecAvg	25.6	1.8	1.5	1.4	1.1	1.2	2.0	3.0	3.7	3.4	2.4	2.0	2.1
(15) Precipitation	PrecMax	33.6	3.5	3.7	3.1	2.7	2.4	3.9	5.2	6.6	6.6	6.9	4.8	4.2
	PrecMin	18.9	0.5	0.2	0.4	0.5	0.6	0.6	1.4	1.2	0.9	0.6	0.2	0.7
	PrecStd	3.9	0.9	0.7	0.7	0.5	0.5	0.7	0.9	1.0	1.7	1.3	1.3	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.4	34.3	41.7	58.7	75.2	82.0	81.8	79.0	65.9	52.0	36.0	30.2
		MCWB	28.6	31.6	35.3	44.5	54.3	60.6	62.0	62.2	51.3	44.6	34.6	27.7
	2%	DB	25.1	29.8	36.3	53.8	70.3	77.4	78.2	72.7	62.7	45.3	31.6	25.0
		MCWB	23.1	27.8	31.0	41.4	52.1	58.0	60.2	59.3	50.2	40.4	30.7	23.4
	5%	DB	20.3	25.5	32.3	49.2	64.4	72.9	74.2	68.3	59.1	41.3	27.1	21.0
		MCWB	19.2	22.6	28.6	39.2	49.2	56.5	58.8	57.6	48.9	37.6	25.3	20.0
	10%	DB	16.3	20.8	28.1	45.5	61.5	70.0	71.8	64.2	55.2	37.5	21.4	16.4
		MCWB	15.4	19.1	24.5	36.7	47.3	55.1	57.8	55.1	47.1	34.8	19.9	15.7
	0.4%	WB	28.5	31.9	35.5	45.3	55.8	63.6	64.4	63.3	54.1	45.4	34.4	28.2
		MCDB	30.0	33.3	41.0	57.1	73.2	78.0	78.7	76.4	63.0	49.6	36.1	30.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	24.1	27.4	31.8	41.5	53.1	60.3	61.8	60.5	52.2	40.9	30.1	24.0
		MCDB	25.8	29.2	35.6	51.6	67.6	72.9	74.1	70.2	60.0	44.9	31.0	25.5
	5%	WB	19.7	23.5	28.5	39.5	50.7	58.3	60.1	58.6	50.3	38.1	25.4	20.6
		MCDB	20.8	26.2	32.2	48.6	62.7	70.1	71.0	66.5	57.2	41.1	27.3	21.5
	10%	WB	16.1	19.6	24.9	37.1	48.7	56.3	58.8	56.6	48.8	35.7	20.8	15.9
		MCDB	16.9	21.5	28.1	45.0	60.2	67.4	68.8	63.1	54.7	38.0	21.9	16.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	14.7	18.1	18.2	18.7	17.3	15.5	14.0	14.3	9.7	11.3	12.4
		MCDBR	14.7	17.4	20.5	22.2	24.5	22.1	21.6	19.8	20.1	12.7	12.5	15.3
		MCWBR	13.6	15.7	16.0	13.3	11.8	10.2	9.4	9.6	11.3	8.9	11.8	14.4
		MCDBR	14.8	17.1	19.2	21.0	22.8	19.6	19.1	17.2	17.5	12.1	12.3	14.7
	5% WB	MCWBR	13.8	15.6	15.2	12.8	11.1	9.5	8.9	9.0	10.1	8.8	11.7	13.9
(40) Clear-Sky Solar Irradiance	taub		0.184	0.246	0.255	0.292	0.338	0.371	0.367	0.388	0.299	0.264	0.186	0.176
	taud		1.983	2.234	2.306	2.317	2.425	2.348	2.403	2.339	2.587	2.468	1.988	1.928
	Ebn at Noon		136	222	272	283	278	268	266	246	258	219	128	59
	Edh at Noon		9	17	25	31	31	35	32	31	20	14	8	4
(44) All-Sky Solar Radiation	RadAvg		66	294	873	1478	1659	1719	1474	1126	732	301	91	24
	RadStd		6	23	59	97	144	119	111	106	95	37	14	2

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+2.02	N/A	+1.28	N/A	N/A	N/A	-688	-783	+121	N/A

Nomenclature: See separate page